C.1 The delivery domain and task

The delivery domain is structured as a grid composed of cells, each containing a specific type of road surface. A task within the delivery domain is illustrated in Figure 3. The state of the delivery agent is its location on the grid. The agent can move one cell in any of the four cardinal directions. Episodes conclude either successfully at the destination, earning a reward of +50, or in failure upon encountering a sheep, resulting in a reward of -50. Non-terminal transitions have a reward equal to the sum of their components: cells with a white road surface carry a -1 reward component, while cells with a brick surface carry a -2 component. Additionally, cells may contain a coin (+1) or a roadblock (-1). Coins remain in place and can, at best, cancel out the cost of the road surface.

Actions that would result in the agent moving into a house or beyond the grid's boundaries result in no movement. In such cases, the reward reflects the current cell's surface component but excludes any coin or roadblock components. The start state distribution, D_0, is uniformly random over non-terminal states.

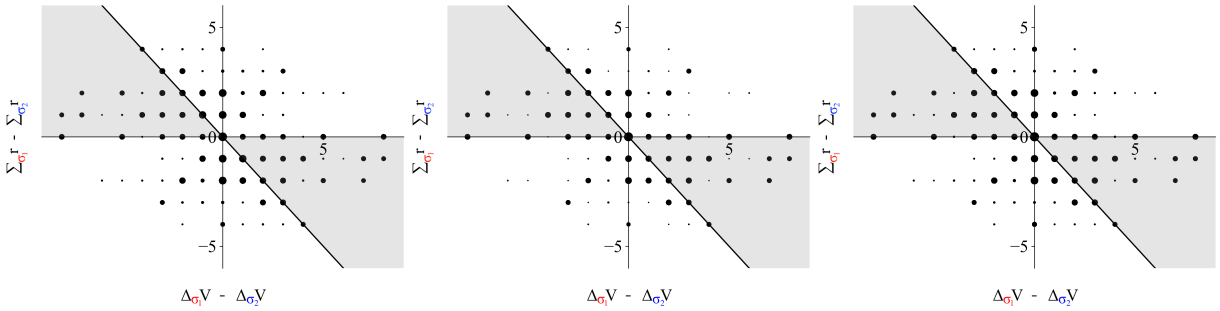
This domain was intentionally designed to make it easy for subjects to recognize poor behavior while making it challenging to discern optimal behavior from most states, mirroring many real-world tasks. This complexity means that some assumptions of the regret preference model, specifically that humans will always prefer optimal segments over suboptimal ones, are not always met, providing a robust test of the model's performance under realistic conditions.

C.2 Selecting segment pairs for preference elicitation

During the main preference elicitation portion of all experimental conditions, preferences are collected over trajectory segments sampled from the delivery task shown in Figure 3. Below we outline our methodology for selecting segment pairs for labeling in the PRIVILEGED experiment, as well as separately for the TRAINED and QUESTION experiments.

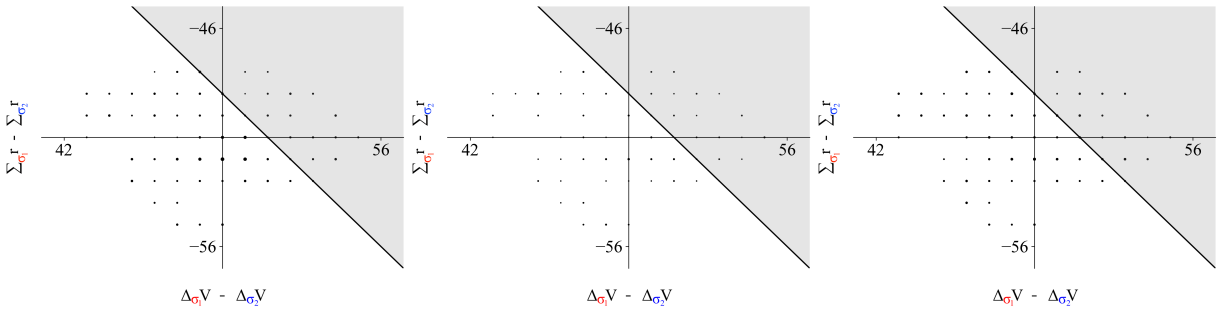
PRIVILEGED Experiment We followed the methodology of Knox et al. (2022) for collecting segment pairs, which involved two stages of data collection with differing goals. The first stage sought to characterize human preferences over a range of possible behaviors, including those that would highlight the differences between partial return and regret. The second stage sought to collect preferences over segment pairs that resolve the identifiablity issues of the partial return preference model related to a constant shift in the reward function. We refer readers to Knox et al. (2022) for a detailed description on how these segment pairs were constructed. Figure 11 plots the coordinates from which segment pairs where sampled for each condition in the first stage of data collection. Figures 12 and 13 plot these coordinates for the second of data collection.

The first stage of data collection resulted in $1,359$ segment pairs from 39 subjects for the P_{Σ_r}-PRIVILEGED condition, $1,418$ segment pairs from 42 subjects for the PRIVILEGED-Control condition, and $1,501$ segment pairs from 43 subjects for the P_{regret}-PRIVILEGED condition. All trajectory segments consisted of 3 actions, and the start state for each segment in a pair was different. The second stage of data collection resulted in $1,173$ segment pairs from 25 subjects for the P_{Σ_r}-PRIVILEGED condition, 375 segment pairs from 8 subjects for the PRIVILEGED-Control condition, and $1,030$ segment pairs from 22 subjects for the P_{regret}-PRIVILEGED condition. For each segment pair in the second stage, the agent in one segment takes 3 actions while in the other segment it reaches a terminal state in fewer than 3 actions. Each subject is asked to label preferences for between 35 and 50 segment pairs. No two subjects see the same segment pairs.



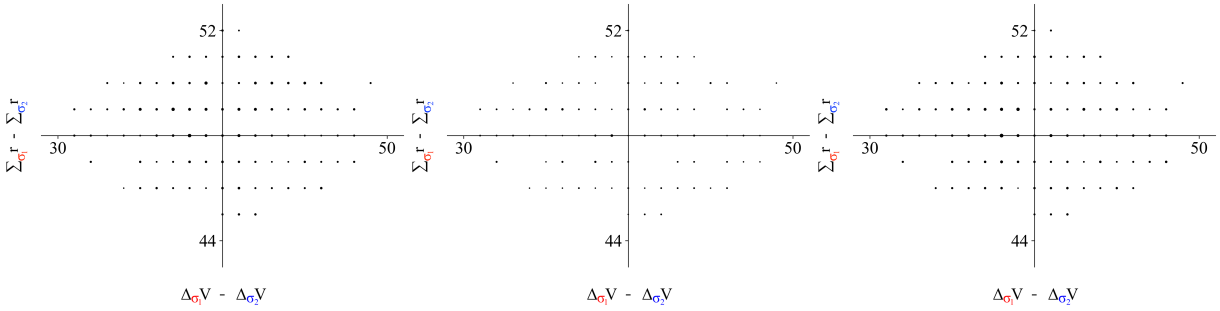
(a) P_{Σ_r}-PRIVILEGED (b) PRIVILEGED-Control (c) P_{regret}-PRIVILEGED

Figure 11: The coordinates of the segment pairs shown to subjects for preference labeling in the first stage of data collection for the PRIVILEGED experiment. The x-axis is the difference in the change in state value between the two segments and the y-axis is partial return differences between the two segments. The areas of the circles are proportional to the number of segment pairs at that point. The proportionality is consistent across this plot and the 3 subplots of Figures 12 and 13.



(a) P_{Σ_r}-PRIVILEGED (b) PRIVILEGED-Control (c) P_{regret}-PRIVILEGED

Figure 12: The coordinates of the segment pairs shown to subjects for preference labeling in the second stage of data collection for the PRIVILEGED experiment. Each segment pair belonging to these graphs contain one segment where the agent terminates at a positive terminal state and one where it does not. The proportionality of the circles are consistent across this plot and the 3 subplots of Figure 11 and 13.



(a) P_{Σ_r}-PRIVILEGED (b) PRIVILEGED-Control (c) P_{regret}-PRIVILEGED

Figure 13: The coordinates of the segment pairs shown to subjects for preference labeling in the second stage of data collection for the PRIVILEGED experiment. Each segment pair belonging to these graphs contain one segment where the agent terminates and one where it does not. The proportionality of the circles are consistent across this plot and the 3 subplots of Figure 11 and 12.

TRAINED and QUESTION Experiments Subjects label the same dataset of 500 segment pairs for each of the 3 conditions in the TRAINED and QUESTION experiment. We chose 500 segment pairs for preference labeling by splitting the P_{Σ_r}-PRIVILEGED, PRIVILEGED-Control, and P_{regret}-PRIVILEGED datasets into different numbers of same sized partitions. We then identified the smallest partition size where the likelihood of the influenced preference dataset was always significantly higher than that of the control condition, defined as being 100 times more likely.

In 72 of the 500 pairs used for labeling, the agent in one trajectory segment takes 3 actions, while in the other segment, it reaches a terminal state in fewer than 3 actions. Knox et al. (2022) found that these trajectory segments are essential for learning a reward function with the partial return preference model in this domain. The remaining trajectory segments consist of 3 actions sampled uniformly randomly from all possible actions. The trajectory segments in a segment pair have the same start state. The order of trajectory segments within a pair and the order of segment pairs shown to subjects is uniformly random. Each subject is asked to label preferences for 50 segment pairs. No two subjects within a condition see the same segment pairs, except for the last segment pair used to test for task comprehension and attention. Figure 19 illustrates the segment pairs sampled for labeling for all experimental conditions within these two experiments.

In the TRAINED experiment we collected data from 10 subjects per condition. In the QUESTION experiment, we sought to do the same but ran into the following issue: We randomly sample subjects to complete our study from a standard sample of available subjects who meet certain criteria (see Appendix C.3), utilize random assignment to assign each subject to a condition, and recollect data removed from subjects who failed the comprehension tests (detailed in Appendix C.4). For the P_{regret}-QUESTION condition, we were unable to find a 10th subject who passed the comprehension test before the subject sample population potentially changed significantly over time. As such, we were no longer confident that we could claim the subjects from all conditions in the QUESTION experiment were drawn from the same population. Therefore, we collected data from only 9 subjects per condition in the QUESTION experiment, resulting in a maximum dataset size of 450 preferences per condition.

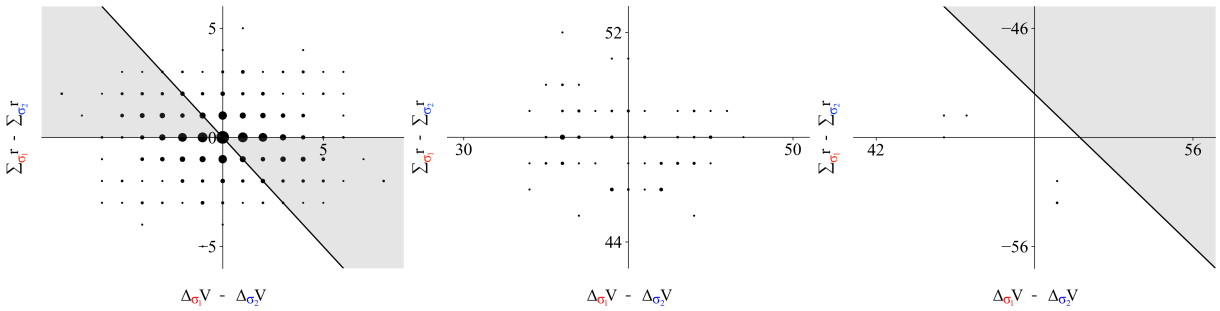


Figure 14: The coordinates of the segment pairs shown to subjects for preference labeling for all conditions for the TRAINED and QUESTION experiments. The areas of the circles are proportional to the number of segment pairs at that point. The proportionality is consistent across this plot.

C.3 Recruiting human subjects

All subject compensation amounts were chosen using the median time subjects took during a pilot study and then calculating the payment to result in $15 USD per hour. This hourly rate of $15 was chosen because it is commonly recommended as an improved US federal minimum wage.

PRIVILEGED Experiment We recruited subjects with IRB approval via Amazon Mechanical Turk and paid subjects $5 per experiment. Subjects had to be located in the United States, have an approval rating of at least 99%, and have completed at least 100 other studies on Mechanical Turk to join our study. Due to an experimental error, we did not show the IRB-approved consent form to participants after they accepted our study on Mechanical Turk. We reported this issue to our IRB and received approval to use the collected data.

TRAINED and QUESTION Experiments We recruited subjects with IRB approval via Prolific. We paid subjects in the TRAINED experiment (see Section 5.2) $7.50 for completing the study. We observed that subjects took about half the time to complete the study when they were not taught about a specific preference model, such as in the control condition, and so for the QUESTION experiment (see Section 5.3) we paid subjects $3.75 for completing the study. Subjects were recruited via Prolific from a standard sample, and were required to be both fluent in English and located in the United States.

C.4 Filtering subject's data

We evaluated each subject's understanding of the delivery domain and excluded those who lacked sufficient understanding. Participants were required to complete a task-comprehension survey, from which we derived a task-comprehension score. The questions and corresponding answer choices are detailed in Table 1. Participants received 1 point for fully correct answers and 0.5 points for partially correct answers. For the PRIVILEGED experiment—and separately for the TRAINED and QUESTION experiments—the threshold score for removing worker data was determined by visually inspecting a histogram of the scores, aiming to strike a balance between upholding high comprehension standards and retaining a sufficient dataset for analysis.

In addition to filtering based on the task-comprehension survey, we also removed data from any participant who ever preferred a segment where the agent ends in a negative terminal state—the worst possible outcome—over a segment where the agent does not.

PRIVILEGED Experiment Subjects could achieve a score on the task-comprehension survey ranging from 0 to 7. Data from participants scoring below 4.5 was discarded. All subjects were shown at least one segment pair containing a segment where the agent ends in a negative terminal state and a segment where it does not, and their data is removed if they prefer the former. These segment pairs are used to test for task comprehension and attentiveness. Because of a data management error, the filtered-out data was lost and we don't otherwise know how many subjects were filtered out for this expirement.

TRAINED and QUESTION Experiment Subjects could achieve a score on the task-comprehension survey ranging from 0 to 6. Data from participants scoring below 3.5 was discarded. The last segment pair shown to participants during preference elicitation always contained a segment where the agent ends in a negative terminal state and a segment where it does not. Other segment pairs shown to subjects may also illustrate this scenario.

Across all conditions in the TRAINED experiment, the data from $19/49$ subjects was removed: $9/19$ from the TRAINED-Control condition, $9/19$ from the P_{Σ_r}-TRAINED condition, and $1/11$ from the P_{regret}-TRAINED condition. The data